KU LEUVEN
DEPARTMENT OF COMPUTER SCIENCE

Patient Monitoring System (PMS)

Part 1

Requirements Analysis

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Attention: Your report still contains 8 instructions or comments. Make sure to delete all of them for the final version. (Run `pdflatex` at least two times after removing them to make sure this warning disappears.)

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H09B5B: Software Architectuur
H07Z9B: Software Architecture

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1. Utility tree of ASRs

TODO: Fill in the table for each ASR.

TODO: For the summary, describe what the requirement is about? Be short but precise.

TODO: For the Business value and architectural impact: shortly explain why.

Note: RequirementType can be any of: {Availability, Interoperability, Modifiability, Performance, Security, Testability, Usability} or some other less-common types: {variability, portability, development distributability, scalability and elasticity, deployability, mobility, and monitorability}

TODO: This table contains a couple of examples for a different system. **Make sure to remove these examples** for your final report.

	Quality Attribute	**Attribute Refinement**	**Summary** **Rationale *Business Value*** **Rationale *Architectural Impact***
1	Interoperability	HIS Integration (HL7 FHIR)	The PMS must integrate with the existing Hospital Information System (HIS) using the HL7 FHIR standard to synchronize patient records and physician workstation functionalities. Condition: Successful exchange of patient records and configuration data between PMS and HIS using FHIR-compliant APIs with 0% data mapping errors.
			H: *This is very important to meet the time constraints of document generation which is the main service DocProc provides to its customers and specified as part of the SLA negotiated with the customers.*
			M: *This requires significant end-to-end performance optimizations at all parts of the document generation process.*
2	Modifiability	Generation of new document types	DocProc wants to offer a new document type without impacting the existing document generation infrastructure. Changes should have **no** impact on the existing document generation infrastructure, but only affect the external elements (templates, raw data, etc.). Extending document processing job initialization should take less than **1** person month to implement, test, and deploy.
			H: *Flexibility in document type offerings is a key enabler for DocProc to adapt and expand its business into the generation and delivery of other documents, thereby providing future opportunities for growth.*
			M: *This requires care in the design and development of the generation infrastructure to ensure it is generic (document-type-agnostic) in its generation capabilities and that document processing jobs are entirely self contained.*
3		TODO	Summary of your ASR of the same quality attribute. This can be a long description that spans multiple lines. The table in this template will automatically wrap the lines. The table can span multiple pages as well, you do not need to create additional tables for this.
			L: *Why does this ASR have a low business value . . .*
			H: *Why does this ASR have a high architectural impact . . .*

2. Quality Attribute Scenarios

Note: In this section, we model the non-functional requirements for the system in the form of *quality attribute scenarios*.
RequirementType can be any of: {Availability, Interoperability, Modifiability, Performance, Security, Testability, Usability} or some other less-common types: { variability, portability, development distributability, scalability and elasticity, deployability, mobility, and monitorability}

TODO: This is an example for a different system. **Make sure to remove this example** for your final report.

2.1 Modifiability: New types of documents

DocProc currently focuses on processing two types of documents: payslips and invoices. However, in the future, DocProc will want to extend the functionality of DocProc to other types of documents as well, for example based on the research of its marketing department. Currently, bank statements seem to be the most promising market segment to focus on next. Therefore, it would be beneficial for DocProc if we could easily extend the system for bank statements as well, and for other kinds of documents afterwards.

- **Source:** The marketing department
- **Stimulus:**
  - Wishes to employ DocProc for other types of documents than payslips or invoices.
- **Artifact:** This modification affects the entire document processing flow: initialization, generation, delivery and storage.
- **Environment:** At design time or at run-time.
- **Response:**
  - This modification should not affect the interface for initializing and initiating document processing jobs, except for the functionality related to checking the raw data for errors.
  - The interface for registering companies should be extended with the functionality to enable the new types of documents for new customer organizations and configure their template for these types of documents and check these templates for errors.
  - The interface for consulting the personal document store and specific documents should be extended with the functionality to show the new types of documents.
  - This modification should not affect the systems that generate documents: generating the new types of documents should reuse as much of the existing functionality for generating documents as possible.
  - Storing generated documents of the new types of documents should not require changes to the existing system for storing generated documents.
- **Response measure:**
  - Extending the interface for initializing document processing jobs takes less than 1 man month to implement, test and deploy.
  - Changing the system to manage templates to support new types of document should take less than 2 man months to implement, test and deploy.
  - Extending the interface for viewing the personal document store and a specific received document takes less than 4 man months to implement, test and deploy.

2.2 RequirementType: Name of the quality attribute scenario

TODO: Shortly describe the context of the scenario.
Next fill in using the structure defined below.

- **Source:** source
- **Stimulus:**
  - Description of a first stimulus.
  - Description of a second stimulus.
- **Artifact:** the stimulated artifact
- **Environment:** the condition under which the stimulus occurs
- **Response:**
  - Describe how the system should respond to the stimulus.
- **Response measure:**
  - Describe how the satisfaction of a response is measured.